DATAPLAY

Gender Prediction Using Naïve Bayes

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Problem Statement

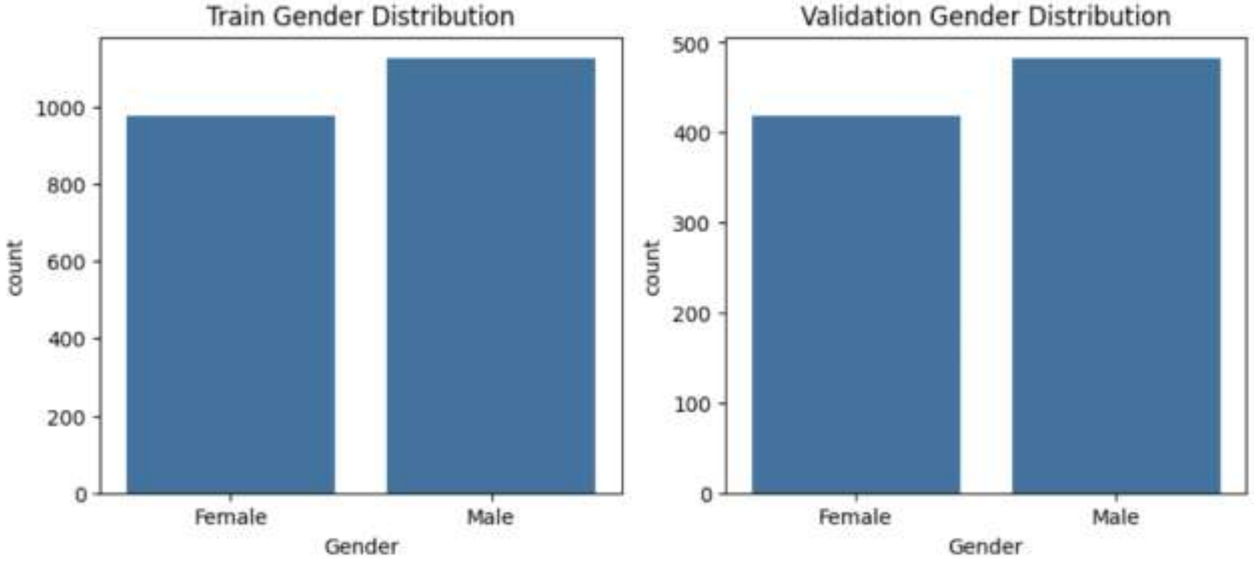
- A dataset with different names of males and females is given. Given a new name, predict the gender of the name.

Feature Engineering

- Extract last letter of the name and predict the gender.
- Extract first and last letter to try for even better results.

Data Collection and Preparation

- We have got a dataset with a total of 3000 entries by DataPlay.
- To build our model, we have done a stratified 30-70 train validation split on the dataset.
- We then check the gender ratio in both train and validation.



Feature Engineering

Do one hot encoding for the last letter column.

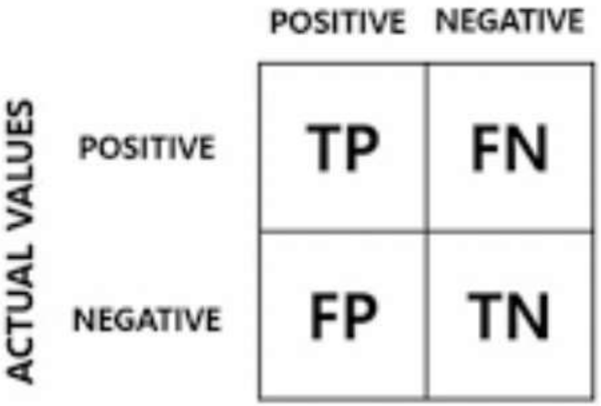
Why one-hot? -> Because last letters are multiple in number and doing a label encoding will put them in unnecessary order for the model.

Apply cross validation and grid search for better results by getting the most optimized hyperparameters.

Model Building (Bernoulli Naïve Bayes)

- P(Male | Kirti) ? P(Female | Kirti)
- P(Male | i) ? P(Female | i)
- P(I | Male)*P(Male) ? P(I | Female)*P(Female)   {P(i) is a constant}
- Naïve Bayes will help us find these values from the dataset to compute the gender given a name.
- Bernoulli because there are binary features representing last letter available with us through one hot encoding.

Model Evaluation



$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Why accuracy is not a good measure always?

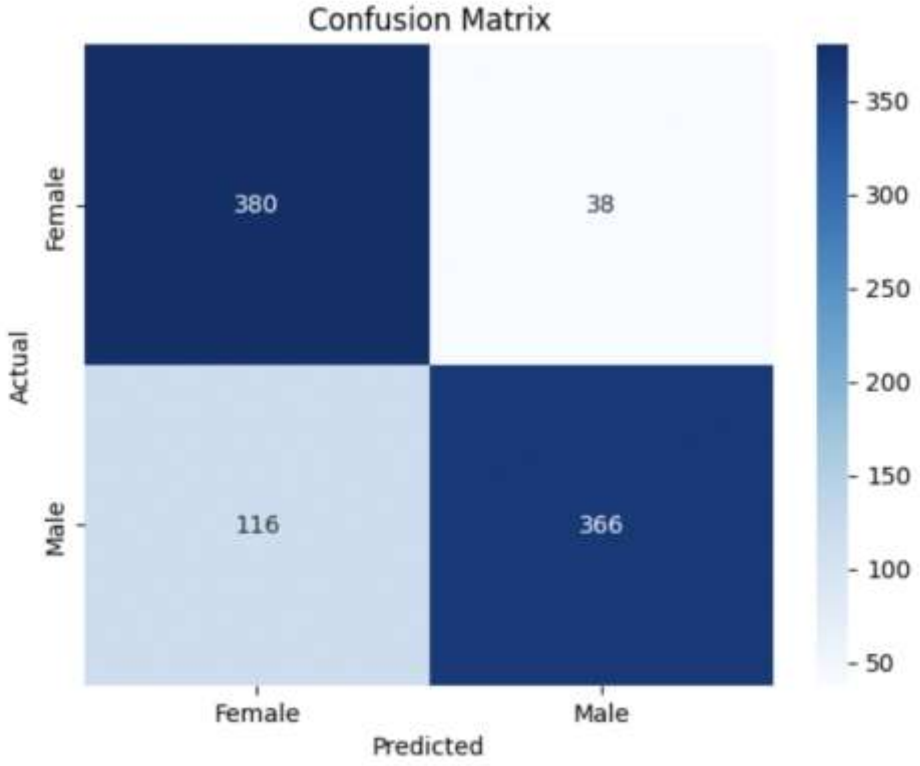
In class imbalance, accuracy gives a wrong indication of the model performance thus averaging away poor recall on minority classes.

Why is F1 score a harmonic mean?

Harmonic mean is dominated by the smaller value, so if either Por R is low, $F_1$is low. This is exactly what we want: you shouldn’t score well if you have high precision but terrible recall (or vice-versa).

Results (using only last letter)

	precision	recall	f1-score	support
Female	0.77	0.91	0.83	418
Male	0.91	0.76	0.83	482
accuracy			0.83	900
macro avg	0.84	0.83	0.83	900
weighted avg	0.84	0.83	0.83	900



Misclassifications done

Total misclassified: 527

	Name	Actual	Predicted
0	Abhicandra	Male	Female
1	Mahajabeen	Female	Male
2	Devadyumna	Male	Female
3	Rupal	Female	Male
4	Acalapati	Male	Female
5	Agneya	Male	Female
6	Punita	Male	Female
7	Mrudu	Female	Male
8	Radhakrishna	Male	Female
9	Kunjabihari	Male	Female
10	Arghya	Male	Female
11	Khushboo	Female	Male
12	Ayeh	Female	Male
13	Jagachandra	Male	Female
14	Ritu	Female	Male
15	Subramani	Male	Female
16	Nithilam	Female	Male
17	Vedavrata	Male	Female
18	Ekalavya	Male	Female
19	Chumban	Female	Male

Results (using first and last letter)

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Best F1 Score (CV): 0.8235163722785185
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